

EC400 Probability and Statistical Inference

Sample Exam 1: Solutions

September 2026

Each question carries 25 marks. The marking guides give an indicative point-by-point allocation; mathematically equivalent valid methods receive full credit.

Question 1

Let $X_1, \dots, X_n$ be iid random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, where $n \geq 2$. Define

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

[25 marks]

(a) Prove the identity

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2.$$

[5 marks]

Solution: Write

$$X_i - \bar{X}_n = (X_i - \mu) - (\bar{X}_n - \mu).$$

Squaring and summing gives

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X}_n - \mu) \sum_{i=1}^n (X_i - \mu) + n(\bar{X}_n - \mu)^2.$$

Since $\sum_i (X_i - \mu) = n(\bar{X}_n - \mu)$, the middle term equals $-2n(\bar{X}_n - \mu)^2$. Combining it with the final term proves the identity.

Marking guide. 1 mark for the decomposition around μ ; 3 marks for the correct expansion and summation; 1 mark for using $\sum_i (X_i - \mu) = n(\bar{X}_n - \mu)$ to obtain the stated identity.

(b) Use part (a) to show that S_n^2 is unbiased for σ^2 . Explain why the denominator is $n - 1$ rather than n . [5 marks]

Solution: For each observation, $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$, so

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 \right] = n\sigma^2.$$

Moreover, independence gives $\text{Var}(\bar{X}_n) = \sigma^2/n$, and $\mathbb{E}[\bar{X}_n] = \mu$. Hence

$$\mathbb{E} [n(\bar{X}_n - \mu)^2] = n \text{Var}(\bar{X}_n) = \sigma^2.$$

Taking expectations in part (a),

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X}_n)^2 \right] = (n-1)\sigma^2, \quad \mathbb{E}[S_n^2] = \sigma^2.$$

Centring at the estimated sample mean uses one degree of freedom. Dividing by $n-1$ exactly corrects the resulting downward expectation of the residual sum of squares.

Marking guide. 1 mark for the expectation of the population-centred sum; 2 marks for evaluating the sample-mean term and obtaining $(n-1)\sigma^2$; 2 marks for concluding unbiasedness and explaining the $n-1$ denominator.

(c) Define the variance estimator with divisor n by

$$V_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Derive its bias. If $v_n = \text{Var}(S_n^2) < \infty$, express the MSEs of V_n and S_n^2 in terms of n , σ^2 and v_n . Explain the bias–variance trade-off. **[6 marks]**

Solution: Since $V_n = ((n-1)/n)S_n^2$, part (b) implies

$$\mathbb{E}[V_n] = \frac{n-1}{n}\sigma^2, \quad \text{Bias}(V_n) = -\frac{\sigma^2}{n}.$$

Moreover,

$$\text{Var}(V_n) = \left(\frac{n-1}{n} \right)^2 v_n.$$

Therefore

$$\text{MSE}(V_n) = \left(\frac{n-1}{n} \right)^2 v_n + \frac{\sigma^4}{n^2}, \quad \text{MSE}(S_n^2) = v_n.$$

The divisor- n estimator introduces downward bias but reduces variance. Which MSE is smaller depends on whether that variance reduction outweighs the squared bias; unbiasedness alone does not settle the MSE comparison.

Marking guide. 1 mark for relating the estimators; 1 for the expectation and bias; 1 for the variance; 2 for the two MSE expressions; 1 for interpreting the trade-off.

(d) Assume additionally that $\mathbb{E}[X_i^4] < \infty$. Starting from the identity

$$S_n^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \right],$$

use the law of large numbers (LLN), the union bound and the probability bound below to prove that $S_n^2 \xrightarrow{p} \sigma^2$. Explain why an estimator can be unbiased without being consistent, or consistent without being unbiased. You may use, for any finite-variance random variable Z and any $t > 0$,

$$\Pr(|Z - \mathbb{E}[Z]| > t) \leq \frac{\text{Var}(Z)}{t^2}.$$

You may also use the following consequence of the displayed identity:

$$|S_n^2 - \sigma^2| \leq \frac{n}{n-1} \left| \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right| + \frac{n}{n-1} (\bar{X}_n - \mu)^2 + \frac{\sigma^2}{n-1}.$$

[9 marks]

Solution: Write

$$A_n = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2, \quad B_n = (\bar{X}_n - \mu)^2, \quad c_n = \frac{n}{n-1}.$$

The fourth-moment assumption ensures that $(X_i - \mu)^2$ has finite variance, so the LLN gives

$$A_n \xrightarrow{p} \mathbb{E}[(X_i - \mu)^2] = \sigma^2.$$

For every $\delta > 0$, the supplied probability bound gives

$$\Pr(B_n > \delta) = \Pr(|\bar{X}_n - \mu| > \sqrt{\delta}) \leq \frac{\sigma^2}{n\delta} \rightarrow 0.$$

Thus $B_n \xrightarrow{p} 0$. Moreover, $c_n \rightarrow 1$, and

$$|S_n^2 - \sigma^2| \leq c_n |A_n - \sigma^2| + c_n B_n + |c_n - 1| \sigma^2.$$

Fix $\varepsilon > 0$. For all sufficiently large n , $c_n \leq 2$ and $\sigma^2/(n-1) < \varepsilon/3$. If both $|A_n - \sigma^2| \leq \varepsilon/6$ and $B_n \leq \varepsilon/6$, the displayed bound is at most ε . Hence the union bound gives

$$\Pr(|S_n^2 - \sigma^2| > \varepsilon) \leq \Pr(|A_n - \sigma^2| > \varepsilon/6) + \Pr(B_n > \varepsilon/6) \rightarrow 0.$$

This proves $S_n^2 \xrightarrow{p} \sigma^2$.

Unbiasedness is a finite-sample expectation statement, whereas consistency is a concentration statement along a sequence of sample sizes. An estimator may be biased at every finite n yet have bias and variance vanish asymptotically; conversely, an unbiased estimator need not become concentrated around its target. Neither property by itself implies the other.

Marking guide. 1 mark for using the supplied identity; 2 marks for the LLN step; 2 marks for applying the supplied probability bound; 2 marks for the final bound and consistency conclusion; 2 marks for distinguishing unbiasedness from consistency.

Question 2

Let D be binary, with $\Pr(D = 1) = p$, where $0 < p < 1$. Conditional on D , a continuous random variable X has density

$$f_{X|D}(x | 1) = 2x, \quad f_{X|D}(x | 0) = 2(1 - x), \quad 0 < x < 1,$$

and both conditional densities are zero otherwise.

[25 marks]

- (a) Verify that the two conditional densities are valid and derive the marginal density $f_X(x)$ over the whole real line. [5 marks]

Solution: Both functions are non-negative on $(0, 1)$, and

$$\int_0^1 2x \, dx = 1, \quad \int_0^1 2(1 - x) \, dx = 1.$$

The mixture formula gives, for $0 < x < 1$,

$$f_X(x) = 2[p x + (1 - p)(1 - x)].$$

Therefore

$$f_X(x) = \begin{cases} 2[p x + (1 - p)(1 - x)], & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Marking guide. 1 mark for non-negativity and integration to one; 1 mark for the mixture formula; 1 mark for the correct expression; 2 marks for stating the support and zero density elsewhere.

- (b) Use Bayes' rule for densities to derive $\Pr(D = 1 | X = x)$ for $0 < x < 1$. Prove that this posterior probability is strictly increasing in x , and interpret the result. [5 marks]

Solution: Bayes' rule gives

$$\begin{aligned} \Pr(D = 1 | X = x) &= \frac{f_{X|D}(x | 1) \Pr(D = 1)}{f_X(x)} \\ &= \frac{px}{px + (1 - p)(1 - x)}. \end{aligned}$$

Its posterior odds are

$$\frac{\Pr(D = 1 | X = x)}{\Pr(D = 0 | X = x)} = \frac{p}{1 - p} \frac{x}{1 - x}.$$

The likelihood ratio $x/(1 - x)$ is strictly increasing on $(0, 1)$, and odds are a strictly increasing transformation of probability. Hence the posterior is strictly increasing in x : larger observations provide stronger evidence for $D = 1$ because their conditional density is relatively larger in that group.

Marking guide. 2 marks for the density version of Bayes' rule; 1 mark for the odds representation; 1 mark for proving monotonicity; 1 mark for the interpretation.

- (c) For an arbitrary $c \in (0, 1)$, derive $\Pr(D = 1 \mid X > c)$ and prove that it exceeds the prior probability p . **[4 marks]**

Solution: The conditional tail probabilities are

$$\Pr(X > c \mid D = 1) = 1 - c^2 = (1 - c)(1 + c),$$

and

$$\Pr(X > c \mid D = 0) = (1 - c)^2.$$

Bayes' rule and cancellation of the positive factor $1 - c$ give

$$\Pr(D = 1 \mid X > c) = \frac{p(1 + c)}{p(1 + c) + (1 - p)(1 - c)}.$$

The likelihood ratio of the event is $(1 + c)/(1 - c) > 1$, so the posterior odds exceed the prior odds and therefore the posterior probability exceeds p .

Marking guide. 1 mark for the two conditional tail probabilities; 1 mark for Bayes' rule and simplification; 2 marks for proving the comparison with the prior.

- (d) Calculate the conditional means and variances of X given each value of D . Then use the laws of iterated expectations and total variance to find $\mathbb{E}[X]$ and $\text{Var}(X)$. **[6 marks]**

Solution: For $D = 1$,

$$\mathbb{E}[X \mid D = 1] = \int_0^1 2x^2 dx = \frac{2}{3}, \quad \mathbb{E}[X^2 \mid D = 1] = \int_0^1 2x^3 dx = \frac{1}{2}.$$

Thus $\text{Var}(X \mid D = 1) = 1/2 - (2/3)^2 = 1/18$. For $D = 0$,

$$\mathbb{E}[X \mid D = 0] = \int_0^1 2x(1 - x) dx = \frac{1}{3},$$

and

$$\mathbb{E}[X^2 \mid D = 0] = \int_0^1 2x^2(1 - x) dx = \frac{1}{6},$$

so $\text{Var}(X \mid D = 0) = 1/6 - (1/3)^2 = 1/18$.

The law of iterated expectations gives

$$\mathbb{E}[X] = p\frac{2}{3} + (1 - p)\frac{1}{3} = \frac{1 + p}{3}.$$

The within-group component of total variance is $1/18$. For the between-group component,

$$\text{Var}(\mathbb{E}[X \mid D]) = p(1 - p) \left(\frac{2}{3} - \frac{1}{3} \right)^2 = \frac{p(1 - p)}{9}.$$

Therefore

$$\text{Var}(X) = \frac{1}{18} + \frac{p(1 - p)}{9}.$$

Marking guide. 2 marks for the two conditional means; 1 mark for the two conditional variances; 1 mark for the unconditional mean; 2 marks for the total variance calculation.

(e) Calculate $\text{Cov}(D, X)$. Are D and X independent? Justify your conclusion.

[5 marks]

Solution: Since D is binary,

$$\mathbb{E}[D] = p, \quad \mathbb{E}[DX] = p \mathbb{E}[X \mid D = 1] = \frac{2p}{3}.$$

Using $\mathbb{E}[X] = (1 + p)/3$ from part (d),

$$\text{Cov}(D, X) = \mathbb{E}[DX] - \mathbb{E}[D]\mathbb{E}[X] = \frac{2p}{3} - \frac{p(1 + p)}{3} = \frac{p(1 - p)}{3} > 0.$$

They are not independent. The nonzero covariance is already sufficient to show this. Equivalently, the posterior probability derived in part (b) varies with x rather than remaining equal to the prior p .

Marking guide. 1 mark for $\mathbb{E}[D]$ and $\mathbb{E}[DX]$; 3 marks for the covariance; 1 mark for the correctly justified independence conclusion.

Question 3

Let X_1, X_2, X_3 be iid $N(\mu, \sigma^2)$ random variables, where $\sigma^2 > 0$, and define

$$A = \frac{X_1 + X_2 + X_3}{3}, \quad B = \frac{X_1 - X_2}{\sqrt{2}}, \quad C = \frac{X_1 + X_2 - 2X_3}{\sqrt{6}}.$$

You may use these facts: a jointly Normal vector with a diagonal covariance matrix has independent entries, and functions of independent groups of variables remain independent. **[25 marks]**

(a) Find the marginal distributions of A, B and C . **[5 marks]**

Solution: Each quantity is a linear combination of independent Normal variables, so each is Normal. Their means are

$$\mathbb{E}[A] = \mu, \quad \mathbb{E}[B] = \mathbb{E}[C] = 0.$$

Independence of the X_i gives

$$\text{Var}(A) = \frac{\sigma^2 + \sigma^2 + \sigma^2}{9} = \frac{\sigma^2}{3},$$

$$\text{Var}(B) = \frac{\sigma^2 + \sigma^2}{2} = \sigma^2, \quad \text{Var}(C) = \frac{\sigma^2 + \sigma^2 + 4\sigma^2}{6} = \sigma^2.$$

Hence

$$A \sim N(\mu, \sigma^2/3), \quad B \sim N(0, \sigma^2), \quad C \sim N(0, \sigma^2).$$

Marking guide. 1 mark for Normality; 1 mark for the three means; 2 marks for the three variances; 1 mark for the complete marginal distributions.

(b) Calculate $\text{Cov}(A, B)$, $\text{Cov}(A, C)$ and $\text{Cov}(B, C)$. Deduce that A, B, C are independent, stating why these three zero covariances are sufficient in this case. **[6 marks]**

Solution: Using the coefficients on the independent X_i ,

$$\text{Cov}(A, B) = \frac{\sigma^2}{3\sqrt{2}}(1 - 1 + 0) = 0,$$

$$\text{Cov}(A, C) = \frac{\sigma^2}{3\sqrt{6}}(1 + 1 - 2) = 0,$$

and

$$\text{Cov}(B, C) = \frac{\sigma^2}{\sqrt{12}}(1 - 1 + 0) = 0.$$

The vector $(A, B, C)'$ is jointly Normal because it is a linear transformation of the jointly Normal vector $(X_1, X_2, X_3)'$. Its covariance matrix is diagonal, so its entries are independent. This conclusion uses joint Normality; zero covariances alone would not imply independence for arbitrary random variables.

Marking guide. 3 marks for the three covariance calculations; 1 mark for recognizing joint Normality; 2 marks for the independence conclusion and its qualification.

- (c) Find the distribution of

$$Q = \frac{B^2 + C^2}{\sigma^2}.$$

[4 marks]

Solution: By parts (a) and (b), B/σ and C/σ are independent standard Normal variables. Therefore

$$Q = \left(\frac{B}{\sigma}\right)^2 + \left(\frac{C}{\sigma}\right)^2 \sim \chi_2^2.$$

Marking guide. 1 mark for standardizing B and C ; 1 mark for using their independence; 2 marks for identifying the χ_2^2 distribution.

- (d) Find the distribution of

$$T = \frac{\sqrt{3}(A - \mu)/\sigma}{\sqrt{Q/2}}.$$

State every distributional and independence fact that you use.

[6 marks]

Solution: The numerator

$$Z = \frac{\sqrt{3}(A - \mu)}{\sigma}$$

is standard Normal because $A \sim N(\mu, \sigma^2/3)$. From part (b), A is independent of (B, C) , so Z is independent of Q , which is a function of (B, C) . Part (c) gives $Q \sim \chi_2^2$. Therefore

$$T = \frac{Z}{\sqrt{Q/2}} \sim t_2$$

by the definition of Student's t distribution.

Marking guide. 1 mark for the standard-Normal numerator; 1 mark for $Q \sim \chi_2^2$; 2 marks for the required independence; 1 mark for matching the defining ratio; 1 mark for the t_2 conclusion.

(e) Define

$$D = \frac{B + C}{\sqrt{2}}, \quad E = \frac{B - C}{\sqrt{2}}.$$

Find their marginal distributions and covariance, and deduce that they are independent. Hence find the distributions of D/E and $(D/E)^2$. **[4 marks]**

Solution: Both D and E are linear combinations of the jointly Normal pair (B, C) , so they are jointly Normal. Parts (a) and (b) give

$$\mathbb{E}[D] = \mathbb{E}[E] = 0, \quad \text{Var}(D) = \text{Var}(E) = \sigma^2,$$

and

$$\text{Cov}(D, E) = \frac{1}{2} \text{Cov}(B + C, B - C) = \frac{1}{2} [\text{Var}(B) - \text{Var}(C)] = 0.$$

Joint Normality and zero covariance therefore imply that D and E are independent $N(0, \sigma^2)$ variables. Consequently,

$$\frac{D}{E} = \frac{D/\sigma}{E/\sigma} \sim \text{Cauchy}(0, 1),$$

as a ratio of independent standard Normal variables. Moreover, D^2/σ^2 and E^2/σ^2 are independent χ_1^2 variables, so

$$\left(\frac{D}{E}\right)^2 = \frac{(D^2/\sigma^2)/1}{(E^2/\sigma^2)/1} \sim F_{1,1}.$$

Marking guide. 1 mark for the two marginal distributions; 1 mark for zero covariance and the joint-Normal independence argument; 1 mark for the Cauchy ratio; 1 mark for the $F_{1,1}$ square.

Question 4

Four pupils are observed in two independently sampled matched pairs, (1, 2) and (3, 4). Within each pair one pupil receives a tutoring programme. Let $z_i = -1$ for control and $z_i = 1$ for treatment, with

$$z = (-1, 1, -1, 1)', \quad y_i = \alpha + \beta z_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon \mid X] = 0.$$

Conditional on the treatment design,

$$\Omega = \text{Var}(\varepsilon \mid X) = \begin{pmatrix} s_1^2 & \rho_1 s_1^2 & 0 & 0 \\ \rho_1 s_1^2 & s_1^2 & 0 & 0 \\ 0 & 0 & s_2^2 & \rho_2 s_2^2 \\ 0 & 0 & \rho_2 s_2^2 & s_2^2 \end{pmatrix}.$$

Here s_1^2, s_2^2 are the error variances in the two pairs, and ρ_1, ρ_2 are the correlations within those pairs. Assume $s_1^2, s_2^2 > 0$ and $-1 < \rho_1, \rho_2 < 1$. **[25 marks]**

- (a) Let $m_0 = \mathbb{E}[y_i \mid z_i = -1]$ and $m_1 = \mathbb{E}[y_i \mid z_i = 1]$. Express m_0, m_1, α and β in terms of one another. **[4 marks]**

Solution: The zero conditional mean restriction gives

$$m_0 = \alpha - \beta, \quad m_1 = \alpha + \beta.$$

Adding and subtracting these equations yields

$$\alpha = \frac{m_1 + m_0}{2}, \quad \beta = \frac{m_1 - m_0}{2}.$$

Thus the treatment-minus-control conditional mean difference is 2β , not β .

Marking guide. 2 marks for the two conditional means; 2 marks for recovering α, β and identifying 2β as the group-mean difference.

- (b) Suppose treatment is randomly assigned within each pair, with each pupil having probability $1/2$ of treatment. Let $D = (z + 1)/2$ and assume $Y = DY(1) + (1 - D)Y(0)$, where $Y(1), Y(0)$ are the potential outcomes. What causal quantity does 2β identify in the experimental population? Does random assignment alone make that population representative of all pupils? **[4 marks]**

Solution: The observed-outcome identity gives $Y = Y(1)$ when $D = 1$ and $Y = Y(0)$ when $D = 0$. Random assignment makes D independent of the potential outcomes conditional on the matching design. Within each pair, the treated and control means therefore identify the corresponding potential-outcome means. Because the treatment probability is $1/2$ in every pair, averaging these comparisons gives

$$2\beta = m_1 - m_0 = \mathbb{E}[Y(1) - Y(0)]$$

for the experimental population represented by the design. Random assignment identifies the effect for these pupils; it does not itself make them a random sample of all pupils. Applying the result to a wider population requires suitable random sampling or additional assumptions.

Marking guide. 1 mark for the observed-outcome identity and random-assignment identifying condition; 1 mark for the ATE interpretation of 2β ; 2 marks for the random-assignment versus representativeness distinction.

(c) Let

$$\bar{y}_0 = \frac{y_1 + y_3}{2}, \quad \bar{y}_1 = \frac{y_2 + y_4}{2}.$$

Using the fact that least-squares fitted values equal the two group sample means in this regression with an intercept and a binary regressor, derive $\hat{\alpha}$ and $\hat{\beta}$. **[5 marks]**

Solution: For the control observations the fitted value is $\hat{\alpha} - \hat{\beta}$, and for treated observations it is $\hat{\alpha} + \hat{\beta}$. Least squares equates these two fitted group means to the observed group means:

$$\hat{\alpha} - \hat{\beta} = \bar{y}_0, \quad \hat{\alpha} + \hat{\beta} = \bar{y}_1.$$

Solving,

$$\hat{\alpha} = \frac{\bar{y}_1 + \bar{y}_0}{2} = \frac{y_1 + y_2 + y_3 + y_4}{4},$$

and

$$\hat{\beta} = \frac{\bar{y}_1 - \bar{y}_0}{2} = \frac{-y_1 + y_2 - y_3 + y_4}{4}.$$

Marking guide. 2 marks for the two fitted-group equations; 1 mark for $\hat{\alpha}$; 2 marks for $\hat{\beta}$.

(d) Describe the heteroscedasticity and dependence represented by Ω . In a large study with many independent matched pairs of this form, which standard-error method would naturally accommodate both features? **[5 marks]**

Solution: Within pair j , both error variances are s_j^2 , their covariance is $\rho_j s_j^2$, and their correlation is ρ_j . Thus $s_1^2 \neq s_2^2$ would represent heteroscedasticity across pairs, while nonzero ρ_j represents within-pair dependence. The zero off-block entries say that the two pairs are uncorrelated. With many independently sampled pairs, standard errors clustered at the matched-pair level naturally allow arbitrary heteroscedasticity and dependence within each pair. White standard errors that assume zero cross-observation covariance would not address the within-pair dependence.

Marking guide. 1 mark for the unequal variances; 2 marks for the within-pair and between-pair covariance pattern; 2 marks for choosing pair-clustered rather than ordinary White standard errors.

- (e) Calculate $\text{Var}(\bar{y}_0 \mid X)$, $\text{Var}(\bar{y}_1 \mid X)$ and $\text{Cov}(\bar{y}_0, \bar{y}_1 \mid X)$. Hence derive the conditional variance of $\hat{\beta}$. Explain how the positive within-pair covariances affect the variance of $\hat{\beta}$ relative to a calculation that incorrectly sets the off-diagonal elements of Ω to zero. **[7 marks]**

Solution : The control mean uses observations 1 and 3, which belong to different pairs, so

$$\text{Var}(\bar{y}_0 \mid X) = \frac{1}{4} [\text{Var}(\varepsilon_1 \mid X) + \text{Var}(\varepsilon_3 \mid X)] = \frac{s_1^2 + s_2^2}{4}.$$

Similarly, $\text{Var}(\bar{y}_1 \mid X) = (s_1^2 + s_2^2)/4$. The two group means share the within-pair covariance terms:

$$\text{Cov}(\bar{y}_0, \bar{y}_1 \mid X) = \frac{1}{4} [\text{Cov}(\varepsilon_1, \varepsilon_2 \mid X) + \text{Cov}(\varepsilon_3, \varepsilon_4 \mid X)] = \frac{\rho_1 s_1^2 + \rho_2 s_2^2}{4}.$$

Using $\hat{\beta} = (\bar{y}_1 - \bar{y}_0)/2$,

$$\text{Var}(\hat{\beta} \mid X) = \frac{(1 - \rho_1)s_1^2 + (1 - \rho_2)s_2^2}{8}.$$

If the within-pair covariances were incorrectly set to zero, the calculated variance would instead be $(s_1^2 + s_2^2)/8$. If both correlations are positive, the covariance reduces the variance of the treated-minus-control difference, so ignoring it overstates, rather than understates, the variance of $\hat{\beta}$. Robust corrections need not always increase a standard error.

Marking guide. 3 marks for the two group-mean variances and their covariance; 3 marks for deriving the variance of $\hat{\beta}$; 1 mark for the correct comparison with the zero-covariance calculation.

Question 5

An education researcher observes m independently sampled classes, with $r \geq 2$ pupils observed in each class. For class $j = 1, \dots, m$ and pupil $k = 1, \dots, r$, suppose

$$Y_{jk} = \mu + A_j + e_{jk},$$

where $A_j \sim N(0, \tau^2)$ are iid class shocks, $e_{jk} \sim N(0, \sigma^2)$ are iid pupil shocks, all class and pupil shocks are independent, and $\tau^2, \sigma^2 > 0$. **[25 marks]**

- (a) Find the mean and variance of Y_{jk} . Find the covariance between two distinct pupils in the same class and between pupils in different classes. Hence calculate the correlation between two pupils in the same class. Recall that correlation is covariance divided by the product of the two standard deviations. **[5 marks]**

Solution: For every pupil,

$$\mathbb{E}[Y_{jk}] = \mu, \quad \text{Var}(Y_{jk}) = \tau^2 + \sigma^2.$$

For $k \neq \ell$, pupils k and ℓ in class j share only the class shock, so

$$\text{Cov}(Y_{jk}, Y_{j\ell}) = \text{Var}(A_j) = \tau^2.$$

Pupils in different classes have independent class and pupil shocks, so their covariance is zero. The correlation between two pupils in the same class is

$$\rho = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Marking guide. 1 mark for the mean; 1 mark for the variance; 1 mark for both covariance cases; 2 marks for the correlation within a class.

- (b) Let

$$\bar{Y} = \frac{1}{mr} \sum_{j=1}^m \sum_{k=1}^r Y_{jk}$$

be the overall sample mean. Express $\bar{Y} - \mu$ in terms of the class and pupil shocks, and derive its exact variance. **[6 marks]**

Solution: Substitution gives

$$\bar{Y} - \mu = \frac{1}{m} \sum_{j=1}^m A_j + \frac{1}{mr} \sum_{j=1}^m \sum_{k=1}^r e_{jk}.$$

The two sums are independent. The class component has variance

$$\text{Var} \left(\frac{1}{m} \sum_{j=1}^m A_j \right) = \frac{\tau^2}{m}.$$

The pupil component has variance

$$\text{Var} \left(\frac{1}{mr} \sum_{j=1}^m \sum_{k=1}^r e_{jk} \right) = \frac{\sigma^2}{mr}.$$

Consequently

$$\text{Var}(\bar{Y}) = \frac{\tau^2}{m} + \frac{\sigma^2}{mr}.$$

Marking guide. 2 marks for the correct decomposition; 3 marks for the two variance components; 1 mark for their sum.

- (c) Use mean-square convergence to show that $\bar{Y}$ is consistent for μ as $m \rightarrow \infty$. **[4 marks]**

Solution: Part (a) gives $\mathbb{E}[\bar{Y}] = \mu$, and part (b) therefore gives

$$\mathbb{E}[(\bar{Y} - \mu)^2] = \text{Var}(\bar{Y}) = \frac{\tau^2 + \sigma^2/r}{m} \rightarrow 0.$$

Thus $\bar{Y} \xrightarrow{m.s.} \mu$. Mean-square convergence implies convergence in probability, so $\bar{Y}$ is consistent for μ .

Marking guide. 2 marks for showing that the mean-squared error tends to zero; 2 marks for the mean-square and consistency conclusions.

- (d) Compare the correct standard error of $\bar{Y}$ with the standard error obtained by incorrectly treating the mr pupil outcomes as iid. Express their ratio in terms of r and the correlation ρ between pupils in the same class. Prove that the iid calculation understates uncertainty and explain the source of the error. **[6 marks]**

Solution: The correct standard error is

$$\text{se}_{\text{correct}}(\bar{Y}) = \sqrt{\frac{\tau^2}{m} + \frac{\sigma^2}{mr}}.$$

Treating all pupils as independent would instead give

$$\text{se}_{\text{iid}}(\bar{Y}) = \sqrt{\frac{\tau^2 + \sigma^2}{mr}}.$$

Their ratio is

$$\frac{\text{se}_{\text{correct}}}{\text{se}_{\text{iid}}} = \sqrt{\frac{r\tau^2 + \sigma^2}{\tau^2 + \sigma^2}} = \sqrt{1 + (r-1)\rho} > 1.$$

The strict inequality follows from $r > 1$ and $\rho > 0$. The iid calculation counts pupils in the same class as if they supplied independent information, although they share A_j ; it omits all positive within-class covariance terms.

Marking guide. 2 marks for the correct and iid standard errors; 1 mark for the ratio; 1 mark for proving the inequality; 2 marks for the dependence interpretation.

- (e) Compare the limits obtained by (i) letting $r \rightarrow \infty$ with m fixed and (ii) letting $m \rightarrow \infty$ with r fixed. In which sequence is the overall sample mean consistent for μ ? Explain the distinction in terms of the independent units in the study. **[4 marks]**

Solution: If m is fixed and $r \rightarrow \infty$, then

$$\text{Var}(\bar{Y}) = \frac{\tau^2}{m} + \frac{\sigma^2}{mr} \rightarrow \frac{\tau^2}{m} > 0.$$

To establish non-consistency, use the decomposition from part (b):

$$\bar{Y} = \mu + \frac{1}{m} \sum_{j=1}^m A_j + \frac{1}{mr} \sum_{j=1}^m \sum_{k=1}^r e_{jk} \xrightarrow{p} \mu + \frac{1}{m} \sum_{j=1}^m A_j.$$

The final error average has mean zero and variance $\sigma^2/(mr) \rightarrow 0$, so it converges to zero in mean square and in probability. The remaining class-shock average is $N(0, \tau^2/m)$: it does not converge to zero as more pupils are added to these same classes. Thus the overall sample mean is not consistent for μ along this sequence.

If r is fixed and $m \rightarrow \infty$, both variance components vanish and part (c) proves consistency. Additional pupils in an existing class share its shock; additional independent classes supply new class-level information.

Marking guide. 1 mark for the non-degenerate common-shock limit when m is fixed; 1 for consistency when $m \rightarrow \infty$; 2 for the independent-cluster interpretation. An argument using the exact Normal distribution is also valid. A positive limiting variance alone is not a general proof of non-consistency.

Question 6

An estimator of $\beta = (\beta_1, \beta_2)'$ is approximately Normal with estimate and estimated covariance matrix

$$\hat{\beta} = \begin{pmatrix} 0.18 \\ -0.36 \end{pmatrix}, \quad \hat{V} = \begin{pmatrix} 0.01 & 0 \\ 0 & 0.04 \end{pmatrix}.$$

For the calculations below, you may use $\Phi(1.80) = 0.9641$, the two-sided 5% standard-Normal critical value 1.96, the 5% critical value 5.991 for χ^2_2 , and $\Pr(\chi^2_2 \geq 6.48) = 0.0392$. You may also use $\sqrt{0.05} \simeq 0.224$. **[25 marks]**

- (a) For the two individual null hypotheses $H_{01} : \beta_1 = 0$ and $H_{02} : \beta_2 = 0$, calculate the z statistics and two-sided p -values. Test each null separately at the unadjusted 5% level. **[5 marks]**

Solution: The estimated standard errors are $\sqrt{0.01} = 0.10$ and $\sqrt{0.04} = 0.20$. Hence

$$z_1 = \frac{0.18}{0.10} = 1.80, \quad z_2 = \frac{-0.36}{0.20} = -1.80.$$

Each two-sided p -value is

$$2[1 - \Phi(1.80)] = 2(1 - 0.9641) = 0.0718.$$

Because $0.0718 > 0.05$, equivalently because $|1.80| < 1.96$, neither individual null is rejected at the unadjusted 5% level.

Marking guide. 1 mark for the standard errors; 2 marks for the two statistics; 1 mark for the two-sided p -values; 1 mark for both decisions.

- (b) Construct the two separate 95% confidence intervals and state whether each contains zero. **[5 marks]**

Solution: The first interval is

$$0.18 \pm 1.96(0.10) = 0.18 \pm 0.196 = [-0.016, 0.376].$$

The second is

$$-0.36 \pm 1.96(0.20) = -0.36 \pm 0.392 = [-0.752, 0.032].$$

Both intervals contain zero.

Marking guide. 2 marks for each confidence interval; 1 mark for stating that both contain zero.

- (c) We now want to test whether *both* coefficients are zero: $H_0 : \beta = \beta_0 = (0, 0)'$. This is a *joint* null hypothesis. A *Wald test* extends the distance-from-the-null idea in Lecture 4 to several coefficients, allowing for their variances and covariances. For this question, use the statistic

$$W = (\hat{\beta} - \beta_0)' \hat{V}^{-1} (\hat{\beta} - \beta_0).$$

You may use the fact that, under this joint null, W is approximately χ^2_2 : there are two restrictions being tested. Calculate W and test at the 5% level, using both the supplied critical value and tail probability. **[6 marks]**

Solution : Since $\hat{V}$ is diagonal,

$$\hat{V}^{-1} = \begin{pmatrix} 100 & 0 \\ 0 & 25 \end{pmatrix}.$$

Therefore

$$\begin{aligned} W &= \frac{0.18^2}{0.01} + \frac{(-0.36)^2}{0.04} \\ &= 1.80^2 + (-1.80)^2 = 6.48. \end{aligned}$$

Under the joint null, the two standardized coordinates are approximately independent standard Normals: the Normal approximation has diagonal covariance. Thus $W = z_1^2 + z_2^2$ is approximately χ^2_2 . Since $6.48 > 5.991$, the null is rejected at 5%. Equivalently, the supplied p -value is $0.0392 < 0.05$, giving the same decision.

Marking guide. 1 mark for inverting the covariance matrix; 2 marks for the Wald calculation; 2 marks for using the approximate reference distribution and critical-value decision; 1 mark for the tail-probability decision.

- (d) Compare the joint decision in part (c) with the separate decisions in part (a). Explain why they can differ. Does rejecting the joint null tell us that each coefficient is nonzero? Does failing to reject an individual null prove that the corresponding coefficient is zero? **[4 marks]**

Solution : Each separate statistic measures one coefficient's distance from zero in standard-error units. The joint Wald statistic adds the two squared statistics: $1.8^2 + (-1.8)^2 = 6.48 > 5.991$, even though $1.8 < 1.96$. Thus the joint statistic exceeds its critical value while neither separate statistic does. Joint rejection means the data provide evidence against both coefficients being zero simultaneously; it does not establish that each coefficient is nonzero. Conversely, failure to reject an individual null is not proof that the corresponding coefficient equals zero.

Marking guide. 2 marks for explaining how the sum of squares exceeds its critical value; 1 mark for the correct meaning of joint rejection; 1 mark for explaining why non-rejection is not proof of a null.

- (e) Using a standard-Normal reference distribution, test each of the following null hypotheses:

$$H_{0a} : 2\beta_1 + \beta_2 = 0, \quad H_{0b} : \beta_1 - \beta_2 = 0.$$

Explain why these decisions need not agree with part (c): what does each null hypothesis ask? **[5 marks]**

Solution: Let $R_a = (2, 1)$ and $R_b = (1, -1)$. Since the estimated covariance is diagonal, the two calculations are

	$R\hat{\beta}$	$R\hat{V}R'$	z	5% decision
H_{0a}	$2(0.18) - 0.36 = 0$	$4(0.01) + 0.04 = 0.08$	0	do not reject
H_{0b}	$0.18 - (-0.36) = 0.54$	$0.01 + 0.04 = 0.05$	$0.54/0.224 = 2.41$	reject

The second decision uses $2.41 > 1.96$; the first statistic has two-sided p -value one. There is no contradiction. Part (c) tests whether both coefficients are zero. The equation $2\beta_1 + \beta_2 = 0$ can hold even if neither coefficient is zero, while H_{0b} asks whether the two coefficients are equal. These are different hypotheses, so their test decisions need not agree.

Marking guide. 1 mark for the first test and decision; 2 marks for the second test's variance calculation, statistic and decision; 2 marks for reconciling both with the joint test.